

Vinh Ly

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Summary

Software Engineer with **18 years** delivering end-to-end, on-call services for high-traffic media and e-commerce platforms. Owned the live-stream recommendation pipeline serving millions of concurrent viewers at YouTube, the real-time auction matching engine at Whatnot, and the global ad-delivery system at Google. Proficient in **Python, Java, C++, Go**, and **Kubernetes**. Cut processing latency by 27% through refactoring core data paths. Partnered with product, data science, and security teams to ship features across continents. Built CI/CD pipelines with **Docker** and **Terraform**, enabling daily deployments across multiple regions. Next focus: driving data-centric product velocity in a remote-first, AI-enhanced CRM environment.

Core Skills

Languages & Scripting: TypeScript, JavaScript, HTML, CSS, SQL, Python

Systems & Networking: AWS

Automation & CI/CD: Ansible, Terraform

Data & Containers: Docker, Kubernetes, Postgres, MongoDB, Redis

Work Experience

Staff Software Engineer, YouTube, 02/2025 - Present (Remote)

- Reduced video thumbnail generation latency by 22% by rewriting the processing service in **Go** and optimizing **Redis** caching, resulting in smoother playback for 15M daily active users across the platform.
- Implemented a fault-tolerant live-chat scaling layer using **Kubernetes** and **AWS** Auto Scaling, handling peak traffic spikes of 3× without service degradation and reduced average response time by 15%.
- Migrated legacy video metadata store to **PostgreSQL** with partitioning, cutting query latency from 120ms to 45ms and saving \$200K in annual storage costs for the global catalog and improved data consistency.
- Designed on-call incident response runbooks and integrated **Prometheus** alerts, decreasing mean time to resolution by 40% across all video services while maintaining SLA compliance for over 100 critical endpoints.
- Collaborated with product and data science teams to launch a recommendation API built on **Python** and **Flask**, driving a 12% increase in user engagement within the first month.

Staff Software Engineer, Whatnot, 05/2024 - 02/2025 (Remote)

- Re-architected the real-time auction matching engine using **Go** and **Redis**, boosting match throughput by 35% and supporting 2× higher concurrent users during peak sale events without latency spikes.
- Implemented a fault-tolerant notification service on **AWS EKS** with **Docker**, achieving 99.99% uptime over six months and reducing alert fatigue by 20% through smarter routing for all user types.
- Optimized the **TaskTiger** scheduler configuration, cutting background job latency by 18% and freeing 15% more CPU capacity for new features across the platform in Q3.
- Led cross-functional on-call rotations, establishing clear escalation paths and reducing mean time to acknowledge incidents from 7 minutes to 3 minutes while maintaining service reliability.

Staff Software Engineer/Manager, Google, 03/2015 - 04/2024 (Remote)

- Directed a team of 12 engineers to build a global ad-delivery platform on **Java** and **Google Cloud Platform**, increasing fill rate by 9% and revenue by \$45M annually across all markets.
- Engineered a high-throughput logging pipeline using **Go**, **Kafka**, and **Elasticsearch**, processing 1.2B events daily and cutting search latency by 40% for real-time monitoring across services.

- Introduced **Terraform** and **Ansible** for infrastructure as code, reducing provisioning time from weeks to hours and cutting operational overhead by 30% while improving consistency.
- Managed on-call rotations and incident postmortems, establishing a blameless culture that lowered repeat incidents by 22% over 2 years and improved team morale.
- Collaborated with product managers to define OKRs, delivering three major feature releases per quarter that increased user retention by 5% across the platform.

Software Engineer, Google, 02/2011 - 02/2015 (Remote)

- Developed core search indexing components in **C++**, improving indexing speed by 28% and supporting a 15% growth in daily queries for the global web service.
- Implemented a distributed caching layer using **Memcached** and **Redis**, reducing average page load time from 1.8s to 1.2s for high-traffic news sites.
- Created internal monitoring dashboards with **Grafana**, enabling engineers to detect anomalies 40% faster and reduce mean time to repair by 30%.
- Contributed to the migration of legacy services to **Google Cloud Platform**, achieving a 25% cost reduction while improving scalability for future growth.

Software Design Engineer II, Microsoft, 10/2007 - 12/2011 (Remote)

- Designed and built a distributed file synchronization service in **C#** and **SQL Server**, supporting 2M daily active users with 99.5% reliability across Windows and Azure platforms.
- Implemented a real-time collaboration feature using **SignalR**, reducing latency from 300ms to 120ms and increasing user satisfaction scores by 14% in the Office suite.
- Optimized database schema and indexes, cutting query execution time by 35% and saving \$500K in hardware costs over 3 years for the enterprise search product.
- Led a cross-team effort to integrate **Active Directory** authentication, improving security compliance and reducing login issues by 22% across all internal tools.

Software Design Engineer Intern, Microsoft, 06/2006 - 08/2006 (Remote)

- Developed a prototype for automated UI testing using **C#** and **Selenium**, decreasing manual test effort by 40% and improving coverage for the Office suite.
- Created a performance benchmark suite in **Python**, identifying bottlenecks that led to a 15% speedup in key modules for the upcoming release.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007